

6	(a)	Number of protons = 82 (start with 88, subtract 2 for every alpha decay and add 1 for every beta decay)	B1	
		Number of neutrons = 126 (Start with 224-88=136; subtract 2 for every alpha decay and subtract 1 for every beta decay)	B1	
	(b)	(i)	Initial number of nuclei = $\frac{12 \times 10^{-3}}{224} \times 6.02 \times 10^{23} = 3.225 \times 10^{19}$	C1
			Decay constant $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{3.63 \times 24 \times 3600} = 2.21 \times 10^{-6} \text{ s}^{-1}$	C1
			Initial activity = $\lambda N = 2.21 \times 10^{-6} \times 3.225 \times 10^{19}$ $= 7.13 \times 10^{13} \text{ s}^{-1}$	M1 A1
		(ii)	Mass of radium-224 nuclei remaining after 6.0 days $m = m_0 \left(\frac{1}{2} \right)^{\frac{t}{t_{1/2}}}$ $m = 12 \left(\frac{1}{2} \right)^{\frac{6.0}{3.63}}$ $= 3.8 \text{ mg}$ also accept methods using decay constant: decay constant for radium-224 = $\lambda = \frac{\ln 2}{3.63} = 0.191 \text{ day}^{-1}$ Mass remaining after 6.0 days $m = 12e^{-0.191 \times 6.0}$ $= 3.8 \text{ mg}$	C1 A1 C1 A1
		(iii)	The <u>half life of Lead-212 is much longer than the half life of Radon-220,</u> and <u>Radon-220 is decaying into Polonium-216 and producing Lead-212 nuclei,</u> thus for these reasons the concentration of Lead-212 is higher than the concentration of Radon-220 at the time specified in the question.	B1 B1

3	(c)	better to have many cables supporting the load instead of relying on one	R2
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